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Original Research

Self-reported physical activity properties and 20-year all-cause and cardiovascular mortality among community-dwelling older adults

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Abstract

Purpose: Physical activity was previously associated with decreased mortality. Current guidelines recommend >150 min/week or >75 min/week of moderate or high-intensity exercise to maintain a healthy lifestyle; however, exercise properties most strongly associated with low mortality among the elderly may still be explored.
Methods: A total of 1210 community-dwelling older adults, from the third phase (1999–2004) of the Israel Study on Glucose Intolerance, Obesity, and Hypertension, were followed until 2016 and 2019 for cardiovascular and all-cause mortality, respectively. Physical activity properties were recorded and evaluated against all-cause and cardiovascular mortality.
Results: Mean age at baseline was 73 ± 7 years, with 638 (53%) females, and 585 (48%) reported habitual exercise. When compared to sedentary individuals, multivariable Cox regressions showed a significantly lower risk for all-cause mortality among currently active individuals [hazard ratio (HR) = 0.72, 95% confidence interval (CI): 0.59–0.88, $P = .002$], those engaging in light-moderate activity (HR = 0.72, 95% CI: 0.57–0.89, $P = .003$), those with diverse exercise types (HR = 0.59, 95% CI: 0.44–0.80, $P = .001$), more sessions/week (HR = 0.94, 95% CI: 0.92–0.97, $P < .001$), those meeting current exercise recommendations (HR = 0.79, 95% CI: 0.58–0.89, $P = .03$), those who engaged in walking (HR = 0.58, 95% CI: 0.45–0.76, $P < .001$), and swimming (HR = 0.66, 95% CI: 0.45–0.96, $P = .03$). Similar HRs were found for cardiovascular mortality, although a somewhat stronger protective association was observed for swimming (HR = 0.48, 95% CI: 0.24–0.95, $P = .04$) compared to a sedentary lifestyle.
Conclusion: The study further supports current exercise guidelines among the elderly. It also underscores the importance of physical activity in older individuals while prioritizing a greater number of sessions/week in addition to the total duration, and highlights specific activity features associated with lower long-term mortality among older adults.

Key messages:

- **What is already known on this topic** – Physical activity was associated with a lower risk for mortality, although the specific properties and the preferred type of exercise among older adults are still debatable.
- **What this study adds** – The study suggests the optimal activity characteristics in older adults while prioritizing activity sessions over time, light-moderate exercise over strenuous activity, diverse activity, and walking and swimming over other activities.
- **How this study might affect research, practice or policy** – Future exercise guidelines should focus on increasing activity sessions throughout the week and not on the cumulative time to maximize the effect on mortality.

Keywords: physical activity; all-cause mortality; cardiovascular mortality; prospective cohort; exercise properties

Introduction

Population aging is one of the major global health issues today and continues to impose a significant burden on health systems around the world. In the USA, the proportion of individuals aged ≥65 has dramatically increased from 4.1% to 12.9% over the past decades [1], and it is expected that by 2030, almost 20% of Americans will be ≥65 years [2].

Healthy aging, defined by the Centers for Disease Control and Prevention as the "development and maintenance of optimal physical, mental and social well-being and function in older adults" [3], and longevity are both targeted by health organizations and governments worldwide and are facilitated through the promotion of a healthy and active lifestyle.

The physical activity guidelines for older adults [4] are fairly similar to those for adults in general, and recommend two to three weekly sessions of muscle-strengthening activities, flexibility, and balance workouts, plus minimum activity levels of 150 weekly minutes of moderate-intensity aerobic exercise, or 75 min/week of vigorous-intensity aerobic exercise. Nevertheless, even lower levels of activity were previously associated with lower mortality [5]. In the USA, the National Health and Nutrition Examination

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Survey 2011–2012 [6] found that 27.3% of older participants (age 65+ years) reported meeting physical activity guidelines, while in Israel, it is estimated that 36% of the older Jewish population and 19% of the older Arab population met the current guidelines' recommendations [7].

Additionally, although physical activity was previously associated with a lower risk for all-cause and cardiovascular (CV) mortality [8], specific properties such as duration per session, frequency (times per week), and the preferred type of exercise among older adults are still debatable. Several studies have shown greater survival with more strenuous activities [9], concerning the cumulative time and intensity, while others showed no survival benefits with intense exercise [5].

We aimed to assess the specific physical activity features associated with lower CV and all-cause mortality among community-dwelling older adults.

Methods

Study population

The study population consisted of community-dwelling older adults who were screened during the third phase (1999–2004) of the Israel study on Glucose intolerance, Obesity, and Hypertension and was previously described elsewhere [10]. This longitudinal study was initiated with 5710 men (50%) and women (50%) randomly sampled from the National Population Registry in 1967, according to origin (based on country of birth: originating in Yemen, Middle East, North Africa, and Europe–North America) and birth decade. Of them, 1719 who participated in phase II, conducted 10 years later (in ∼1980), and who survived, were invited to participate in phase III. Of them, 1214 consented and were interviewed regarding their health status, physical activity habits, and medication use. Anthropometric measurements (i.e. weight, height, hip and waist circumference) and blood tests were completed for 982 participants.

Blood tests were analyzed by a single laboratory. All participants signed an informed consent to be included in the study prior to the examination. The protocol was approved by the Sheba Medical Center's IRB.

Study endpoints and covariates

Individuals with information on physical activity habits ($n = 1210$) were included in the final analysis.

The primary outcome was all-cause mortality, and the secondary outcome was CV mortality, retrieved from the Israeli Central Population Registry. Follow-up began on the date of examination and ended on the date of death or December 2019 for all-cause mortality, whichever came first, and on the date of CV death or December 2016 for CV mortality, whichever came first.

Physical activity properties were obtained using a structured questionnaire and categorized into characteristics such as current physical activity status (exercising or not at all), level of intensity [light-moderate exercise, defined as household activities (going up the stairs, working in the garden), walking, yoga, dancing, Feldenkrais, aerobics, and physiotherapy-associated activity; and intense exercise, defined as running, cycling, swimming, ball games (tennis, soccer, volleyball, basketball, bowling), and gym], exercise diversity (one vs. more than one type of exercise), adherence according to years of past physical activity, total duration per week (min), frequency (sessions per week), meeting current exercise guidelines target of ≥150 min/week of moderate-intensity or ≥75 min/week of vigorous-intensity activities [4], and type of specific activity including walking, exercise at home, swimming, cycling, and light sport [defined as yoga, Feldenkrais, dancing, and household activities (gardening, climbing stairs)].

Glycemic status was defined according to the American Diabetes Association criteria [11]: normoglycemic—fasting glucose <100 mg/dl or 2-h post oral glucose tolerance test (OGTT) plasma glucose <140 mg/dl and no reports of diabetes at baseline or of using anti-diabetes medications; pre-diabetes—fasting plasma glucose 100–125 mg/dl or 2-h post-OGTT plasma glucose 140–199 mg/dl and no reports of diabetes at baseline or using anti-diabetes medications; diabetes—fasting plasma glucose >125 mg/dl, or 2-h post-OGTT plasma glucose >199 mg/dl, or reports of the presence of diabetes or of using anti-diabetes medications.

Statistical analysis

Differences between activity status and participant's characteristics were assessed using Student's *t*-test or the Mann–Whitney nonparametric test, when appropriate, for continuous variables. Categorical variables were compared using the chi-square test or Fisher's exact test for small cells. The multivariable Cox proportional hazards model was applied to evaluate the association between physical activity properties and mortality outcomes, presented by hazard ratio (HR) with a 95% confidence interval (95% CI). The models were adjusted for available known CV and all-cause mortality risk factors and included age, sex, ethnic background according to country of birth, smoking status, systolic blood pressure, glomerular filtration rate (GFR; low defined as GFR <60 ml/min according to the Cockcroft and Gault formula), total cholesterol, body mass index (BMI) groups (normal <25 kg/m^2, overweight 25–29.9 kg/m^2, obese BMI ≥30 kg/m^2), glycemic status, history of myocardial infarction (MI), 10-year past hospitalizations, and physical activity properties. Each physical activity property was implemented in a separate model, adjusting for the above covariates: model 1—current physical activity versus sedentary; model 2—exercise intensity; model 3—exercise diversity; model 4—exercise adherence, i.e. number of years of past physical activity; model 5—exercise duration per week (min); model 6—exercise frequency, i.e. number of sessions per week/-categorized into three groups (1–3, 4–6, and ≥7 times per week); model 7—type of specific activity.

Multicollinearity between the study variables was tested and covariates showing high correlation (Spearman's rank correlation coefficient or the Pearson correlation ≥60%) were not introduced into the same multivariable model. Variables that were found to be significantly associated with mortality in the univariate model were included in the multivariable model.

The proportional hazards assumption for the multivariable Cox regression model was tested using the log minus log plot.

To avoid reverse causality, a sensitivity analysis was conducted by excluding subjects who died within 2 years from baseline, i.e. with a follow-up duration of <2 years.

All analyses were performed using SPSS software (version 25) and Python software (version 3.10). Statistical significance was defined as a two-sided *P* value of ≤.05.

Results

Mean baseline age was 73 ± 7 years, and 638 (53%) were females (Table 1).

Almost half of the cohort, 585 (48%), were currently physically active, for a median past adherence of 10 [interquartile range (IQR) 31] years. In total, 285 subjects (24%) reached the current exercise guidelines target. Those who were currently active reported a median weekly exercise duration of 120 [IQR 120] min per week,

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Table 1. Baseline characteristics of the study cohort ($n = 1210$) according to current physical activity status.

	Sedentary ($n = 625$)	Physically active ($n = 585$)	P value	Total ($n = 1210$)
Age (years), mean ± SD	74 ± 8	72 ± 7	<.001	73 ± 7
Female, *n* (%)	355 (57)	283 (48)	.03	638 (53)
Origin, *n* (%)			<.001	
Europe–America	177 (28)	244 (42)		421 (35)
Middle East	180 (29)	139 (24)		319 (26)
Yemen	156 (25)	106 (18)		262 (22)
North Africa	112 (18)	96 (16)		208 (17)
Smoking			.04	
Never, *n* (%)	374 (60)	328 (56)		702 (58)
Past, *n* (%)	180 (29)	213 (36)		393 (33)
Current, *n* (%)	71 (11)	44 (9)		115 (10)
Pack years, median [IQR]	30 [40]	21 [33]	.03	24 [35]
Specific diet[a], *n* (%)	225 (36)	267 (46)	.001	492 (41)
BMI (kg/m^2), mean ± SD, $n = 1041$	28.9 ± 5	27.5 ± 4	<.001	28 ± 5
Normal, *n* (%)[b]	116 (23)	134 (23)	<.001	250 (24)
Overweight, *n* (%)[b]	194 (38)	286 (49)		480 (46)
Obese, *n* (%)[b]	196 (39)	115 (20)		311 (30)
Missing	119 (19)	50 (9)		169 (14)
Waist circumference (cm), mean ± SD	100.6 ± 13	97.7 ± 20	<.001	99.1 ± 20
Male, mean ± SD	101.6 ± 12	99.7 ± 10	.06	100.6 ± 11
Female, mean ± SD	99.7 ± 13	95.3 ± 11	<.001	97.6 ± 11
Systolic blood pressure (mmHg), mean ± SD, $n = 1018$	139.4 ± 20	134.8 ± 21	<.001	137 ± 20
Diastolic blood pressure (mmHg), mean ± SD, $n = 1019$	78.9 ± 11	76.2 ± 11	<.001	78 ± 11
Albumin (g/dl), mean ± SD, $n = 962$	4.4 ± 0.3	4.4 ± 0.3	.2	4.4 ± 0.3
Total protein (g/dl), mean ± SD, $n = 962$	7.2 ± 0.5	7.1 ± 0.5	<.001	7.2 ± 0.5
Total cholesterol (mg/dl), mean ± SD, $n = 986$	196.8 ± 40	194.8 ± 39	.4	196 ± 39
HDL (mg/dl), mean ± SD, $n = 986$	49.4 ± 13	49.4 ± 13	.9	49.4 ± 14
Triglycerides (mg/dl), median [IQR], $n = 976$	119 [76]	116 [74]	.2	118 [75]
LDL (mg/dl), mean ± SD, $n = 965$	119 ± 31	119 ± 31	.9	119 ± 31
Fasting glucose (mg/dl), mean ± SD, $n = 982$	116 ± 44	112 ± 32	.1	113 ± 38
Fasting insulin (mU/L), median [IQR], $n = 976$	10.6 [10]	9.7 [8]	.01	10.1 [8.6]
Glycemic state[c]			.6	
Normoglycemia, *n* (%)	210 (42)	229 (45)		439 (43)
Pre-diabetes, *n* (%)	102 (20)	97 (19)		199 (20)
Diabetes, *n* (%)	191 (38)	182 (36)		373 (37)
GFR (ml/min), mean ± SD	66 ± 20	66 ± 18	.1	66 ± 19
Low GFR[d], *n* (%)	192 (42)	189 (39)	.5	381 (40)
Baseline medications				
Statins, *n* (%), $n = 737$	109 (30)	129 (35)	.1	238 (20)
Anti-diabetic, *n* (%), $n = 737$	65 (18)	68 (18)	.4	133 (11)
Anti-hypertensive, *n* (%), $n = 748$	199 (54)	178 (47)	.08	377 (31)
Years of past physical activity, median [IQR]	10 [31]			10 [31]
Exercise, duration per week (min), median [IQR]		120 [120]		120 [120]
Reported comorbidities and risk factors				
Polypharmacy[e], *n* (%)	62 (10)	53 (9)	.6	115 (10)
Hypertension, *n* (%)	300 (48)	230 (39)	.01	530 (44)
S/P MI, *n* (%)	69 (11)	67 (12)	.4	136 (11)
S/P CVA, *n* (%)	69 (11)	31 (5)	<.001	100 (8)
VTE, *n* (%)	32 (5)	24 (4)	.3	56 (5)
Lung disease, *n* (%)	35 (10)	20 (6)	.04	55 (8)
10-year hospitalization, *n* (%)	406 (65)	349 (60)	.1	755 (63)

[a]Specific diet: those who reported consuming a specific diet including low fat, low calories, low sugar, low salt, low protein, vegan/vegetarian, and other. [b]BMI categories: normal, <25 kg/m^2; overweight, 25–29.9 kg/m^2; obese, BMI ≥30 kg/m^2. [c]Glycemic state classification: normoglycemic—fasting glucose <100 mg/dl or 2-h post-OGTT (oral glucose tolerance test) plasma glucose <140 mg/dl and no reports of having diabetes or using anti-diabetes medications; pre-diabetes—fasting plasma glucose 100–125 mg/dl or 2-h post-OGTT plasma glucose 140–199 mg/dl and no reports of having diabetes or using anti-diabetes medications; diabetes—fasting plasma glucose >125 mg/dl or 2-h post-OGTT plasma glucose >199 mg/dl or reports of having diabetes or using anti-diabetes medications. [d]Low GFR—defined as GFR <60 (ml/min) according to the Cockcroft and Gault formula. [e]Polypharmacy—defined as ≥5 different prescription medications BMI, body mass index; GFR, glomerular filtration rate; MI, myocardial infraction; CVA, cerebrovascular accident; LDL, low-density lipoprotein; VTE, venous thromboembolism

with a median frequency of 5 [IQR 4] sessions per week, and 225 (19%) reported combining more than one type of exercise. The most common combination was walking + swimming, followed by walking + light activity, and walking + gym. Moreover, 415 (34%) were engaged in light-moderate physical activity and 170 (14%) in intense physical activity (not shown).

Exercise included walking [285 (49%)], exercise at home [89 (15%)], swimming [111 (19%)], gym [21 (4%)], running [7 (1%)], cycling [19 (3%)], light sport [41 (7%)], and ball games [12 (2%)].

Compared with physically active participants, individuals with a sedentary lifestyle were older (74 ± 8 vs. 72 ± 7 years, $P < .001$), with a female predominance (57% vs. 48%, $P = .03$), and had

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a greater waist circumference (100.6 ± 20 cm vs. 97.7 ± 20 cm, $P < .001$).

All-cause mortality

Until December 2019, 613 (50.7%) all-cause deaths were recorded, and 14 374 person-years were accrued, with a median follow-up duration of 13 [IQR 8] years.

Multivariable Cox model (Table 2, Fig. 1A) showed a 30% lower risk for all-cause mortality among those who were physically active at baseline (HR = 0.72, 95% CI: 0.59–0.88) compared with sedentary individuals.

Light-moderate activity (model 2) was significantly associated with reduced mortality (HR = 0.72, 95% CI: 0.57–0.89) compared with sedentary individuals. More intense exercise was associated with a borderline significance of 30% lower mortality risk (HR = 0.74, 95% CI: 0.54–1.0).

Similarly, those engaging in only one type of exercise (model 3) had a 21% lower risk for mortality (HR = 0.79, 95% CI: 0.64–0.99), while those who were engaged in 2 or more activities had a 41% lower risk for mortality (HR = 0.59, 95% CI: 0.44–0.80), *P* for trend <.001.

Furthermore, every 1-unit increment in activity sessions per week was associated with a 6% lower all-cause mortality (HR = 0.94, 95% CI: 0.92–0.97). When specific cutoffs were implemented (model 6), a dose–response relationship was noted, with a 23% (HR = 0.77, 95% CI: 0.59–0.99) lower risk for mortality with 1–3 sessions/week, 31% (HR = 0.69, 95% CI: 0.53–0.92) with 4–6 sessions/week, and 41% (HR = 0.59, 95% CI: 0.44–0.78) with ≥7 sessions/week compared with sedentary individuals (*P* for trend <.001).

Walking (model 7) was associated with a 42% lower risk for mortality (HR = 0.58, 95% CI: 0.45–0.76). This association was also observed for swimming (HR = 0.66, 95% CI: 0.45–0.96) but not for other types of sports.

Years of past physical activity and minutes of exercise per week (as a continuous variable) were not significantly associated with mortality, although meeting current exercise guidelines of at least 150 min/week of aerobic exercise was associated with a 21% decreased mortality risk (HR = 0.79, 95% CI: 0.58–0.89).

The sensitivity analysis performed by excluding individuals with a follow-up duration of <2 years to mitigate reverse causation ($n = 67$) yielded similar results.

Cardiovascular mortality

During the follow-up period, 240 (20%) individuals died from CV causes, with a median follow-up of 10 [IQR 6] years.

Multivariable Cox regression (Table 2, Fig. 1B) demonstrated a significant lower risk for CV mortality among currently active participants (model 1) (HR = 0.66, 95% CI: 0.48–0.92), engaging in light-moderate activities (model 2) (HR = 0.68, 95% CI: 0.47–0.97), and with more than one type of exercise (model 3) (HR = 0.54, 95% CI: 0.33–0.89, *P* for trend = .01). More sessions/week, were associated with lower CV mortality as well (HR = 0.93, 95% CI: 0.88–0.97), although only >6 times per week reached statistical significance (HR = 0.46, 95% CI: 0.27–0.75) (*P* for trend = .001).

Walking and swimming were associated with a lower risk for CV mortality (HR = 0.52, 95% CI: 0.34–0.81 and HR = 0.48, 95% CI: 0.24–0.95, respectively), an association that was not demonstrated in other types of sports. Exercise duration per week (as continuous variable in minutes) was associated with only borderline significant survival benefits (HR = 0.99, 95% CI: 0.99–1.0), but not for those who met current activity recommendations (HR = 0.82, 95% CI: 0.52–1.22).

Discussion

In the current analysis of 1210 community-dwelling older adults with a median follow-up of 13 years, physical activity was associated with a 30% lower risk for all-cause and CV mortality. Moreover, lower CV and all-cause mortality were observed among individuals engaging in light-moderate exercise and more diverse activity.

We detected a dose–response relationship with exercise frequency, such that with increasing activity sessions, lower all-cause and CV mortality were observed, while this was only partially demonstrated for the cumulative minutes of activity per week and for meeting current exercise recommendations. Additionally, walking and swimming were associated with a 30% and 50% lower risk for CV and all-cause mortality.

Intensity and specific activity

While the benefits of exercise in reducing all-cause and CV mortality are well established among middle-aged and younger adults, in the elderly population, this effect is controversial [12] and could benefit from additional data to recommend a more accurate exercise regimen specifically tailored to this vulnerable population. A systematic review with meta-analysis [13] with 28 523 participants aged ≥65 years found lower mortality risk and better physical function among active participants. In another large study with 22 709 participants aged ≥60 years, exercise significantly reduced the risk of falls [Relative Risk (RR) = 0.88; 95% CI: 0.79–0.98] but not hospitalization or mortality [12].

With respect to exercise intensity, several studies have shown that individuals engaging in more strenuous exercise had a lower mortality risk than those engaging in light-moderate exercise, as demonstrated by a dose–response association, whereas other studies showed a U-shaped association with mortality [8,14,15][p17].

Watts *et al.* [14] showed that individuals who reported practicing higher-intensity activities at a high frequency had a greater mortality risk than those who reported practicing moderate activity. In the Copenhagen City Heart Study of 1098 joggers and 413 sedentary non-joggers aged 20 to 93, Schnohr *et al.* [16] found that moderate joggers had a greater mortality risk compared with light joggers, and strenuous joggers had a similar mortality risk as sedentary non-joggers.

In individuals >85 years old, Osawa *et al.* [5] showed that a light amount of exercise was associated with a lower risk of mortality, whereas more strenuous exercise did not reduce the risk of death.

Concerning the type of activity, walking was found to be associated with decreased CV risk factors and lower mortality [17]. Several studies have shown decreased blood pressure [18], lower CV and cerebrovascular risk [19], and all-cause mortality reduction [19] with walking, supporting the promotion of walking as the preferred type of exercise in adults.

As for swimming, the association with lower mortality and CV risk remains controversial [20, 21]. A cohort study with 40 547 men aged 20–90 years found that swimmers had lower mortality rates compared to sedentary individuals, walkers, or runners [22]. Other studies emphasized specific features of swimming such as being well tolerated among patients with severe heart failure [23]. Several mechanisms could explain the observed associations. First, maintaining light-moderate exercise habits may increase the adherence to an active lifestyle, thus increasing the likelihood of persisting throughout aging, functional decline, and sickness. In our cohort, those who were engaged in light-moderate activity had more prolonged years of activity compared with those

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Table 2. Cox regression multivariable models, 1–7, for the associations between sports-related physical activity properties and all-cause and cardiovascular mortality.

Characteristic	Reference category/unit increments	13-year all-cause mortality ($n = 613$)		10-year cardiovascular mortality ($n = 240$)	
		HR	95% CI	HR	95% CI
Sports-related physical activity, current (model 1)*	Never/past	0.72	0.59, 0.88	0.66	0.48, 0.92
Age	1 year	1.10	1.08, 1.11	1.09	1.07, 1.12
Sex, male	Female	0.88	0.7, 1.09	0.96	0.67, 1.38
Origin	Yemen				
Middle East		0.94	0.71, 1.24	1.07	0.68, 1.69
North Africa		0.91	0.66, 1.24	1.21	0.72, 2.02
Europe–America		0.84	0.65, 1.11	1.01	0.64, 1.59
Smoking status	Never				
Past		0.99	0.79, 1.24	0.96	0.67, 1.37
Current		1.16	0.82, 1.64	1.56	0.92, 2.66
Systolic blood pressure	5 mmHg	1.04	1.0, 1.01	1.02	1.0, 1.02
BMI groups[a]	Normal				
Overweight		0.78	0.61, 1.01	0.67	0.45, 1.01
Obese		1.01	0.75, 1.35	0.95	0.59, 1.49
Glycemic state[b]	Normoglycemia				
Pre-diabetes		1.0	0.77, 1.31	0.69	0.43, 1.11
Diabetes		1.36	1.09, 1.72	1.51	1.06, 2.15
Low GFR[c]	Normal GFR	1.19	0.93, 1.52	1.48	0.99, 2.22
Total cholesterol	10 mg/dl	0.99	0.99, 1.0	0.99	0.99, 1.03
Past MI	No	1.74	1.31, 2.32	2.44	1.62, 3.68
10-year hospitalization	None	1.31	1.06, 1.62	1.53	1.07, 2.19
Physical activity, intensity (model 2)*	Never/past				
Light-moderate[d]		0.72	0.57, 0.89	0.68	0.47, 0.97
Vigorous[e]		0.74	0.54, 1.0	0.61	0.35, 1.03
Physical activity, diversity (model 3)*	Never/past				
One type of sport		0.79	0.64, 0.99	0.71	0.49, 1.03
More than one type		0.59	0.44, 0.80	0.54	0.33, 0.89
Years of past physical activity (model 4)*	1 year	0.99	0.97, 1.0	1.03	0.92, 1.15
Time duration per week (min) (model 5)*	1 min	0.99	0.9, 1.0	0.99	0.99, 1.0
Meeting current exercise guidelines[f]	Not	0.79	0.58, 0.89	0.82	0.52, 1.22
Total number of sessions per week (model 6)*	1 unit	0.94	0.92, 0.97	0.93	0.88, 0.97
1–3	None	0.77	0.59, 0.99	0.69	0.47, 1.06
4–6		0.69	0.53, 0.92	0.79	0.51, 1.22
≥7		0.59	0.44, 0.78	0.46	0.27, 0.76
Specific activity (model 7)*	Never/past				
Walking ($n = 285$, 49%)		0.58	0.45, 0.76	0.52	0.34, 0.81
Exercise at home ($n = 89$, 15%)		1.12	0.81, 1.56	1.22	0.73, 2.05
Swimming ($n = 111$, 19%)		0.66	0.45, 0.96	0.48	0.24, 0.95
Cycling ($n = 19$, 3%)		1.01	0.52, 1.97	0.74	0.24, 2.47
Light sports[g] ($n = 41$, 7%)		0.67	0.43, 1.29	0.62	0.16, 1.49

*All models were adjusted for age, sex, origin, smoking, systolic blood pressure, BMI, glycemic status, GFR, total cholesterol, past MI, and 10-year past hospitalizations [a]BMI categories: normal, <25 kg/m^2; overweight, 25–29.9 kg/m^2; obese, BMI ≥30 kg/m^2 [b]Glycemic state classification: normoglycemic—fasting glucose <100 mg/dl or 2-h post-OGTT (oral glucose tolerance test) plasma glucose <140 mg/dl and no reports of having diabetes or using anti-diabetes medications; pre-diabetes—fasting plasma glucose 100–125 mg/dl or 2-h post-OGTT plasma glucose 140–199 mg/dl and no reports of having diabetes or using anti-diabetes medications; diabetes—fasting plasma glucose >125 mg/dl or 2-h post-OGTT plasma glucose >199 mg/dl or reports of having diabetes or using anti-diabetes medications [c]Low GFR—defined as GFR <60 (ml/min) according to the Cockcroft and Gault formula [d]Light-moderate exercise: household activities (going up the stairs, working in the garden), walking, yoga, dancing, Feldenkrais, aerobics, and physiotherapy [e]Vigorous: running, cycling, swimming, ball games (tennis, soccer, volleyball, basketball, bowling), and gym [f]Meeting the current exercise guidelines target of ≥150 min/week of moderate-intensity or ≥75 min/week of vigorous-intensity activities [g]Light sports: yoga, Feldenkrais, dancing, household activities (gardening, climbing stairs) HR, hazard ratio; CI, confidence interval; GFR, glomerular filtration rate; MI, myocardial infraction; BMI, body mass index

engaged in strenuous activity (not shown). Second, intense activity, specifically endurance training, was previously associated with an increased risk for arrhythmias, ventricular remodeling, and coronary calcification, hence increasing the risk for morbidity and mortality in this population [24, 25]. Although this phenomenon was described in younger athletes, perhaps older adults may have a lower threshold for cardiac remodeling. Third, walking can be incorporated into the daily routine, thus contributing to its implementation in daily activities while improving adherence to exercise [17]. Lastly, swimming includes both components of aerobic and anaerobic activity, as well as improving balance, joint mobility, and coordination, without straining the spine and joints, thus contributing to a better functional status and adherence to physical exercise in older age [26].

Exercise frequency and duration

Despite the current American College of Sports Medicine and the American Heart Association physical activity recommendations [27], several studies have shown mortality benefits with a minimum of 30–59 min/week of running (5 to 10 min/day) [28] or 15 min a day or 90 min a week of moderate-intensity exercise [29].

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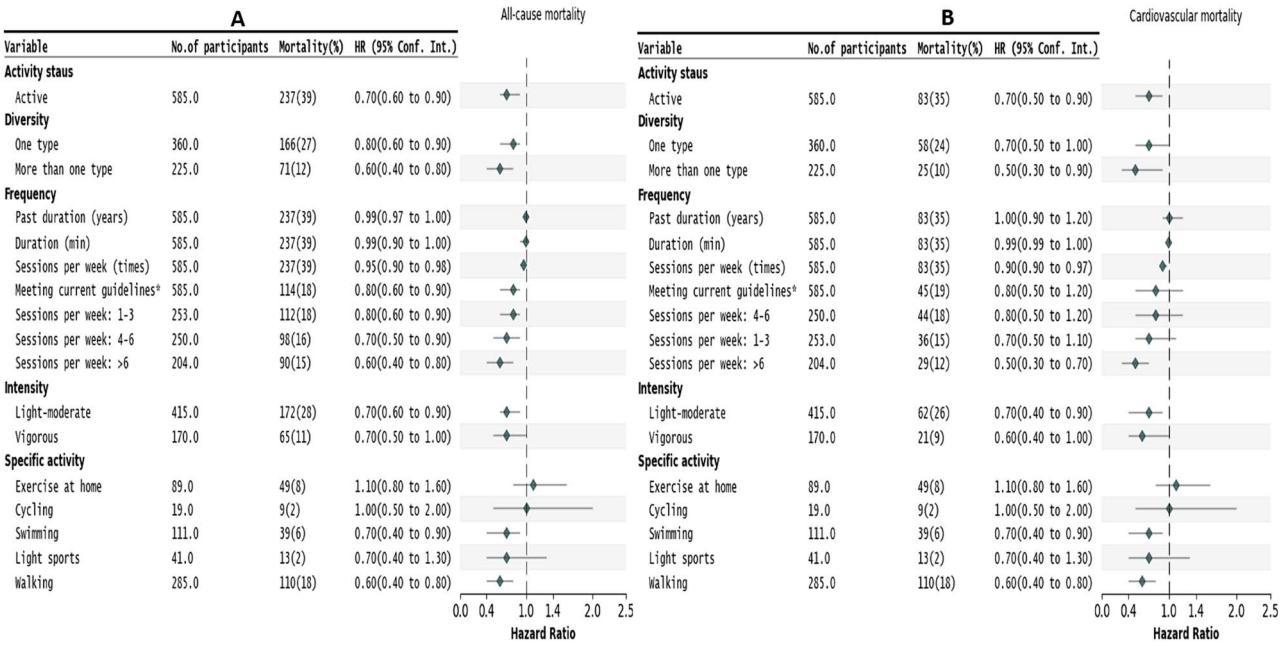


Figure 1 The association between physical exercise properties and all-cause and cardiovascular mortality of community-dwelling elderly men and women. Multivariable [a]Cox proportional hazards analysis. Forest plot describing the association between physical exercise properties and (A) all-cause and (B) cardiovascular mortality. [a]Adjusted for age, sex, ethnicity (originating in Yemen, Asia, North Africa, and Europe/North America), current smoking, systolic blood pressure, low GFR [defined as GFR < 60 (ml/min) according to the Cockcroft and Gault formula], total cholesterol (mg/dl), and BMI groups (normal, <25 kg/m^2; overweight, 25–29.9 kg/m^2; obese, BMI ≥ 30 kg/m^2). Glycemic state classification: normoglycemic—fasting glucose <100 mg/dl or 2-h post-OGTT (oral glucose tolerance test) plasma glucose <140 mg/dl and no reports of having diabetes or using anti-diabetes medications; pre-diabetes—fasting plasma glucose 100–125 mg/dl or 2-h post-OGTT plasma glucose 140–199 mg/dl and no reports of having diabetes or using anti-diabetes medications; diabetes—fasting plasma glucose >125 mg/dl or 2-h post-OGTT plasma glucose >199 mg/dl or reports of having diabetes or using anti-diabetes medications, history of myocardial infarction (MI), 10-year hospitalization, and exercise properties. ∗Current guidelines: ≥150 min/week of moderate-intensity or ≥75 min/week of vigorous-intensity activities

Schnohr *et al.* [30] found among 7308 healthy adults aged 20–93 that walking intensity rather than duration was associated with decreased all-cause mortality. In the Copenhagen City Heart Study, Schnohr *et al.* [16] showed that the optimal frequency of jogging was ≤3 times/week, while a frequency >3 was not more beneficial; however, this subgroup was relatively small.

In our cohort, the lower mortality benefit associated with exercise was dose dependent, such that for every additional weekly activity session, a 5–10% decrement in all-cause and CV mortality was observed, with the lowest mortality risk achieved with ≥7 activity sessions per week. Furthermore, overall weekly exercise duration was associated with only a modest 0.01% CV mortality reduction with each 1-min increment of exercise (95% CI: 0.99–1.0), but not with all-cause mortality (HR = 0.99, 95% CI: 0.99–1.02), while meeting current activity guidelines had a 20% lower risk for all-cause mortality (HR = 0.79, 95% CI: 0.58–0.89) but did not reach statistical significance with CV mortality (HR = 0.82, 95% CI: 0.52–1.22).

In contrast to current guidelines [27], and previous studies in younger adults that prioritized the total duration of exercise/week and even supported similar survival benefits among individuals who practice intense physical activity during the weekend (i.e. "weekend warriors") [31], the current study emphasizes that the number of sessions and not the cumulative duration is a better target to reduce mortality, similar to previous guidelines that highlight the importance of reducing sedentary behavior through daily life while practicing light activity over strenuous exercise [32]. Moreover, in our cohort only 285 (24%) participants reached the current exercise guidelines target, similar to the prevalence in the USA [6], although slightly lower than the prevalence in the general older Israeli Jewish population [7]. These findings should encourage health providers to further implement current exercise guidelines in this population.

Additionally, our findings support a more diverse exercise regimen, as demonstrated by a 40% and 50% reduced risk for all-cause and CV mortality, respectively, among those who engaged in more than one type of exercise, similar to current exercise guidelines [33].

Our results, within the limitations of being an observational study, suggest that the optimal exercise regimen for older adults should include diverse, light-to-moderate activity, preferably walking or swimming, and more sessions throughout the week and not the total exercise duration as recommended by current guidelines, for reduced mortality. These associations should be further evaluate using interventional studies.

Strengths and limitations

The current study has several limitations. First, this study includes the more resilient and relatively young subjects who survived to that point, thus increasing the risk for survival bias, although this study correctly represents older adults living in the community.

Second, similar to many other surveys [7, 34] this study is based on self-reported information, and no objective measurements of physical activity were used. The data for this study were collected >20 years ago when wearables were not as prevalent and accessible as nowadays. Concurrent collection of objective data to quantify physical activity information on a large elderly cohort would be more feasible today. This limitation may increase the risk of information bias, mainly that of recall bias.

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Third, due to the relatively small sample size, we were underpowered to examine the association in some of the subgroups, potentially resulting in misleading conclusions regarding the associations for specific activities.

Fourth, we do not have data regarding the reliability and validity of the self-reported questionnaire, although the physical activity section in the questionnaire was constructed by a sports physician to include several important aspects of exercise.

Lastly, as an observational study, causality cannot be inferred.

Nonetheless, the current analysis presents several important advantages, such as the representation of community-dwelling older adults of an ethnically diverse cohort, detailed data on baseline characteristics such as pharmacologic treatment, and comorbidities, comprehensive analysis of exercise features, long follow-up duration, and extensive adjustments for covariates in the multivariable model while including sensitivity analysis excluding events occurring in the first 2 years of follow-up.

Conclusions

Light-moderate exercise, walking and swimming, more activity sessions per week, and diverse exercise were associated with lower all-cause and CV mortality risk in older adults. Older adults should be encouraged to engage in light-moderate and diverse exercise, for the majority of the week, to reduce mortality. Future exercise guidelines for older adults should focus on increasing activity sessions throughout the week rather than the cumulative time to maximize the effect on survival, in contrast to the current guidelines recommendations.

Research questions

- What is the preferred type of exercise among older adults for better health and longevity?
- What is the minimal amount of exercise required for decreased mortality in the elderly?
- Is increased exercise frequency throughout the week a better target than the cumulative duration for maintaining a healthy lifestyle and decreased mortality?

Main messages

- Specific exercise properties such as light-moderate exercise over strenuous activity, walking and swimming, more frequent activity sessions per week over the total duration, and combining diverse types of exercise were associated with lower all-cause and CV mortality risk in older adults.
- Future exercise guidelines for older adults should focus on increasing activity sessions throughout the week and not on the cumulative time to maximize the effect on mortality.
- Optimal exercise regimen for older adults should include light-to-moderate activity, preferably walking, including combinations of exercise with increasing number of sessions throughout the week for greater longevity.

Author contributions

Y.M. contributed to the data analysis, interpretation of data, and drafting of the manuscript. R.D. and A.C. contributed to the acquisition of the data, conception and design of the work, data analysis, and drafting of the manuscript. A.C. and R.D. critically revised the manuscript. All gave final approval and agreed to be accountable for all aspects of the work, ensuring integrity and accuracy.

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Data availability

Data including the total database will be fully disclosed upon reasonable request to the corresponding author.

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